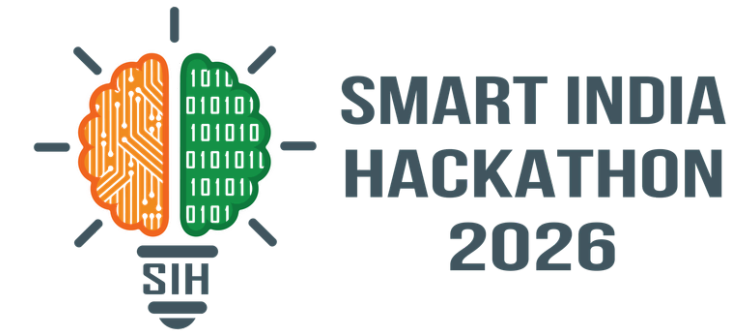
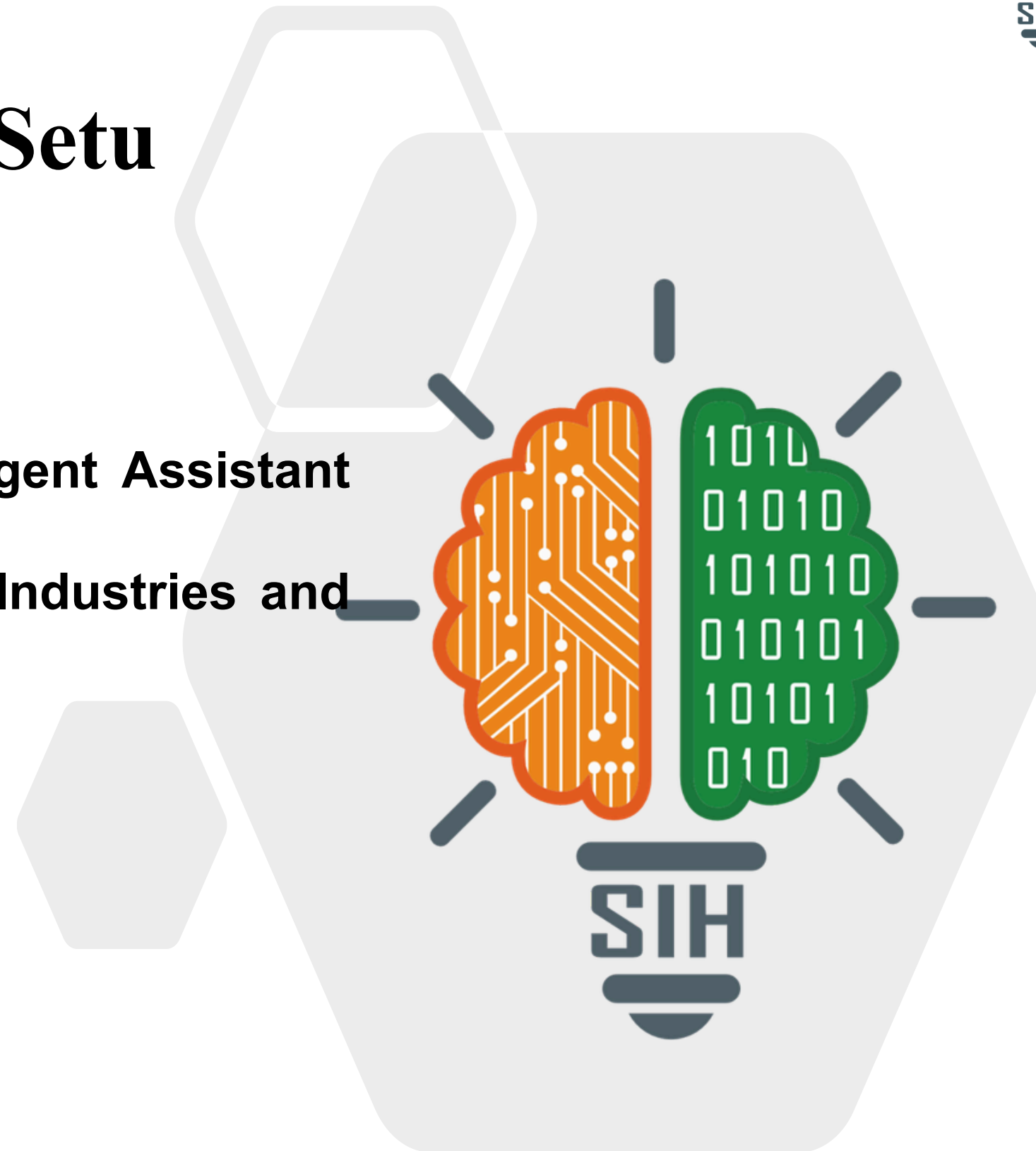


SMART INDIA HACKATHON 2026



BIS-Setu

- Problem Statement ID – SIH26107
- Problem Statement Title- AI-powered Intelligent Assistant for Indian Standards and BIS Services for Industries and Consumers
- Theme- Smart Automation
- PS Category- Software
- Team ID-
- Team Name - NextGen Innovators



BIS-Setu

PROPOSED SOLUTION



AI Compliance Assistant:

A smart AI app that helps businesses understand Indian standards, find the right requirements, and complete compliance step by step.

HOW IT HELPS



Simple Guidance

Explains complex standards in easy language and gives clear next steps.



Smart Search

Quickly finds relevant Indian standards and requirements.



Step-by-Step Checklist

Shows what documents, tests, and approvals are needed.



Cost & Timeline

Gives an easy view of expected tests, approvals, costs, and timelines.



Smart Alerts

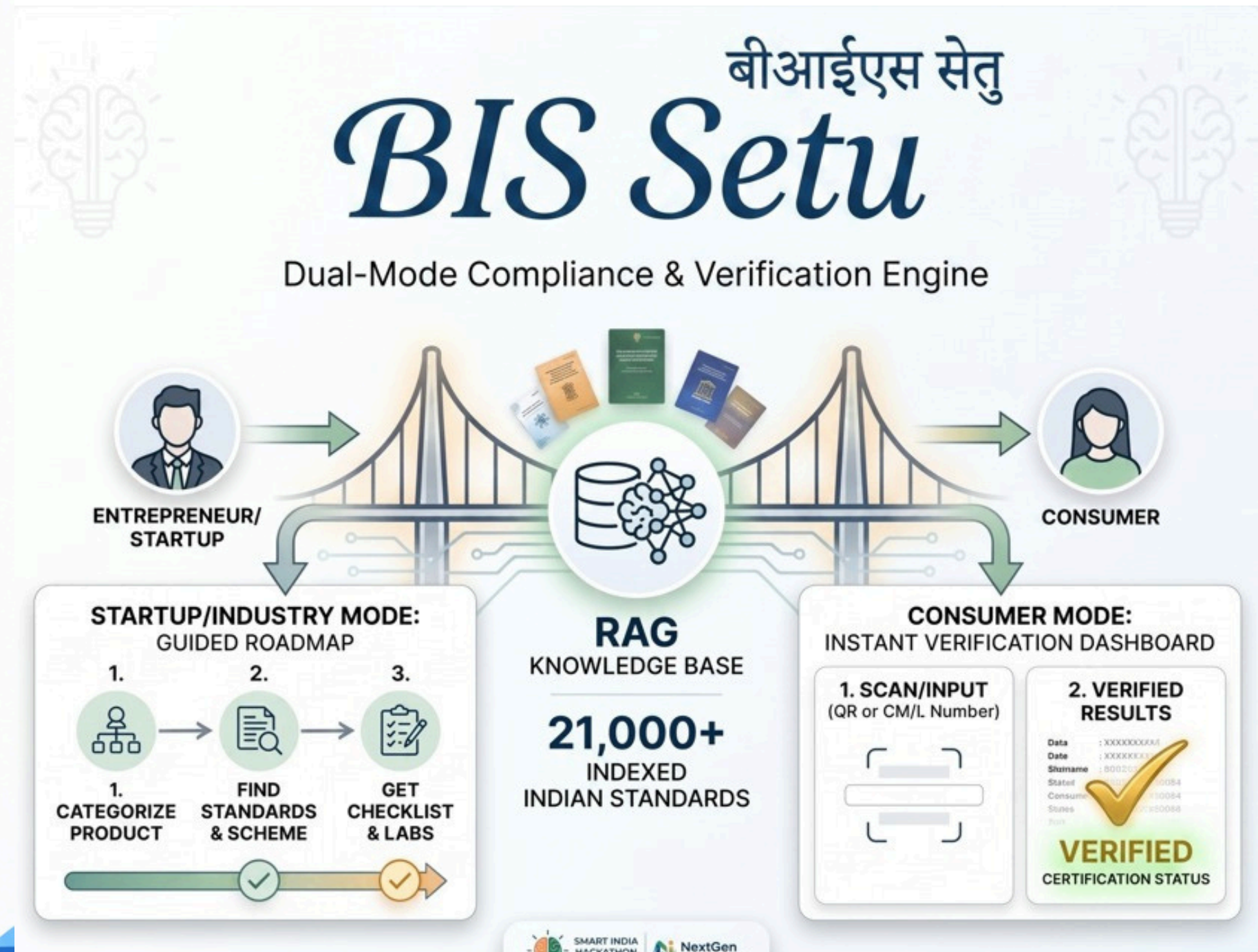
Reminds businesses about certificates, licences, and renewals.

WHAT MAKES IT UNIQUE



AI + Verified Standards

Combines AI guidance with a trusted standards knowledge base.



Frontend & Interface



Backend & Cloud Infrastructure

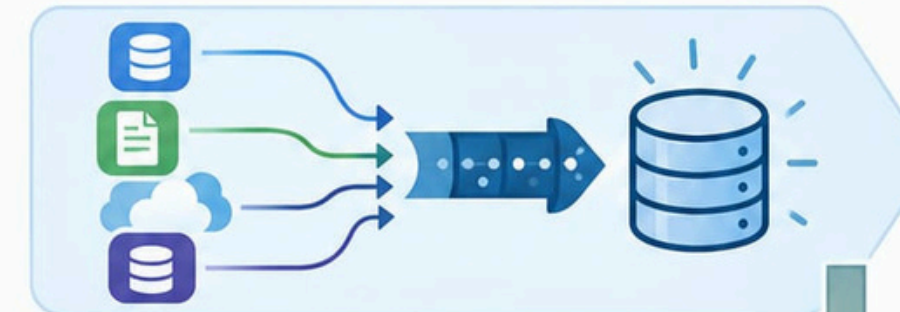


Google Cloud
Platform (GCP)

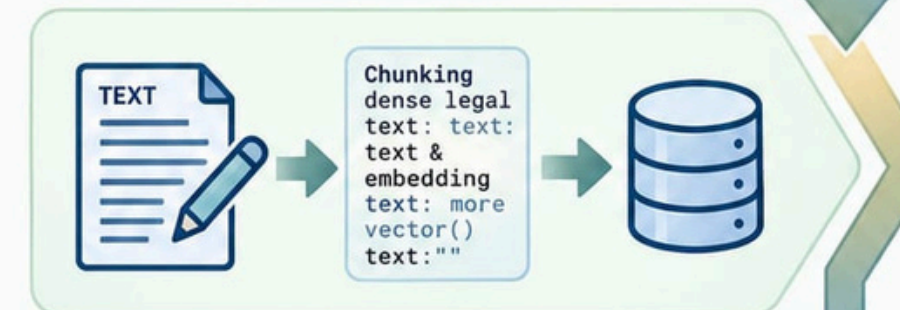
AI Engine & Data Store



BIS Setu: METHODOLOGY & IMPLEMENTATION PIPELINE



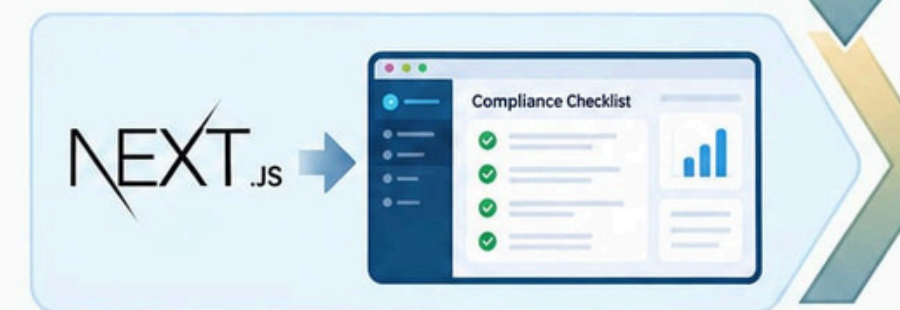
1. Backend Data Ingestion Pipeline
Our backend data ingestion pipeline collects and processes 21,000+ Indian Standard documents



2. PROCESSING & VECTORIZATION
Chunking dense legal text & embedding into Vectors
Vector Store (e.g., ChromaDB or Pinecone)



3. AI RETRIEVAL (RAG)
Match User Queries (Products/Licenses) with precise Indian Standards
LangChain / Llamaindex Orchestration



4. DASHBOARD OUTPUT
Deliver actionable Compliance Checklists & Roadmaps
Serve to React/Next.js Dual-Mode Frontend

Technical Feasibility



- Agent Development Kit framework
- Autonomous AI-driven RAG workflows
- Precise standard chunk retrieval
- Zero-hallucination

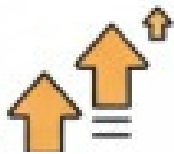
Operational Feasibility



Process & Operational Readiness

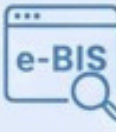
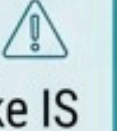
- React / Next.js collaborative stack
- Rapid deployment & live preview workflows
- Seamless continuous integration
- Intuitive, dual-mode interface for end users

Scalability



- Google Cloud Platform backend deployment
- High-throughput vector database scaling
- Zero initial deployment overhead
- Enterprise-grade architecture for heavy query traffic

Defining the Difference

Focus Area	Status Quo (The Challenge)	With BIS Setu (The Strategy)
 Challenge e-BIS Scraper Tool • Data Ingestion 	 Risk • Manual scraping required, fragile data feeds, missing documents 	 Pre-index PDFs (CRS) • Pre-indexed database of ~21,000+ PDFs from e-BIS. Stable, reliable data source 
 Generative LLM (Thinking) • AI Hallucination	 Wrong IS Standard numbers quoted • Prone to hallucinating fake IS standard numbers 	 Strict Zero-Gen RAG Constraint • Strict Zero-Generation RAG constraints. • Summarizes only retrieved clauses. • Exact Citations
 MSMEs (Entrepreneur Group) • Industry Adoption	 Risk  • MSMEs doubt third-party tool guidance	 Verification (Official Stamp/Link) • Direct Verified Official Links to Gazette & e-BIS portals. • 100% Verifiability

Target Audience Impact

MSMEs & Early-Stage Innovators:



Accelerates Go-To-Market: -- Removes compliance guesswork for new product proposals and hardware prototypes.

Streamlines Pre-Incubation: Allows founders to integrate

regulatory standards from day one of product design.

Reduces Cost: Eliminates the need for expensive third-party legal consultants to interpret complex schemes.

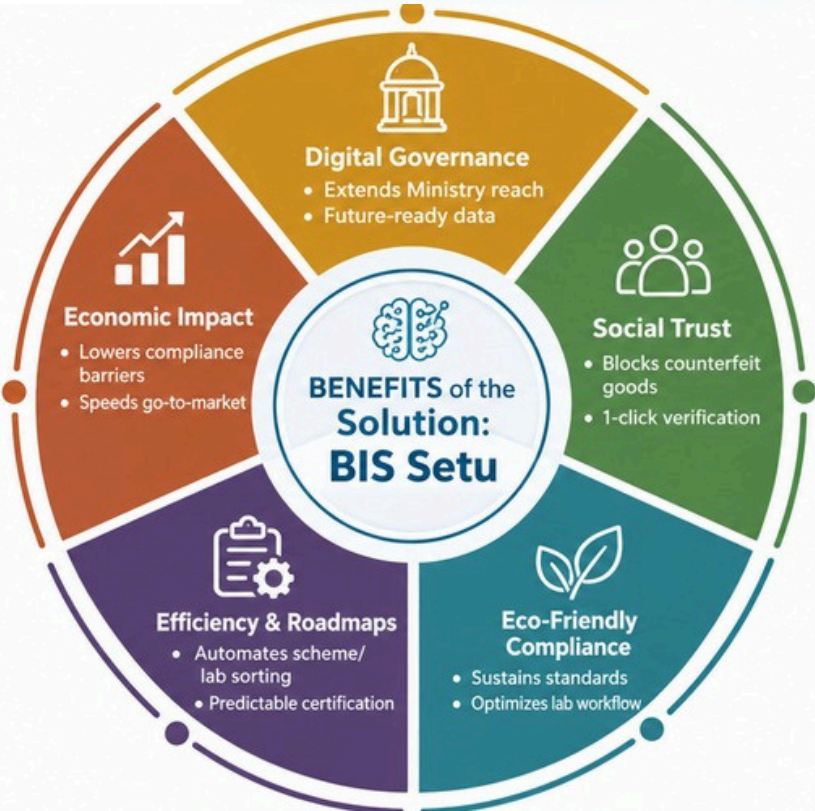
Everyday Consumers



- **Instant Protection:** Frictionless, 1-click verification against counterfeit or unsafe goods.
- **Simplified Safety:** Translates dense government safety ratings into a clear, understandable "Verified" status.
- **Empowered Purchasing:** Fosters a culture of confident and informed consumerism.

BIS Setu: Defining the Core Benefits (Comparison)

Aspect	1 Economic Growth	2 Social Trust	3 Digital Governance
 Key Focus	• Accelerates compliance and market access • Supports startup ecosystem	• Builds consumer confidence • Ensures fair and transparent market practices	• Enhances efficiency of government processes • Reduces manual intervention
 Main Benefit	• Lower compliance barriers • Boosts innovation and investment	• Protects consumer rights • Promotes fair competition	• Simplifies access to information • Improves transparency and accountability
 Key Stakeholders	• Businesses & Startups • Manufacturers	• Consumers • Industry & Market Players	• Ministry / Government • Regulatory Bodies
 Outcome	• Stronger industry ecosystem • Sustainable economic development	• Greater confidence • Safer and reliable products	• Efficient, transparent and future-ready governance • Better public service delivery



Rooted in data integrity and powered by insight, our platform transforms raw information into a space of clarity, resilience, and prosperity—where every public institution becomes an informed decision-maker, every sector a pillar of a digitally advanced future.

Core References

-  **Bureau of Indian Standards Official Portal (bis.gov.in):**
Reference point for existing "Know Your Standard" features and consumer search parameters.
-  **e-BIS Standardization Portal:**
Primary data repository for the open-source extraction of the ~21,000 free Indian Standard (IS) PDFs.
-  **BIS Care Application:**
Baseline reference used to map out current consumer-facing verification frameworks (CM/L and CRS data flows).

Gemini Nano Banana-

Used for internal image generation and visual prototyping (Tool Reference – Gemini Nano)

Future Scope & Conclusion

- **Phase 2 Vision:**
Integration with the **ONDC (Open Network for Digital Commerce)** to automatically verify product safety ratings for e-commerce listings.
- **Multilingual Support:**
Expanding the RAG AI explanation layer to support regional Indian languages via